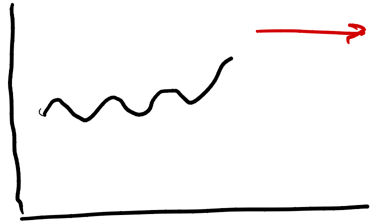
Forecasting:

we are using past obs
from a time series
to make predictions
about the future



Conceptually, if we are thinking about
how this applies in business.

1 Paradigm 1: Forecasting based on your
forecasting horizon

short-term
forecast.

mid-term forecast

next month or
quarter

Long-term

> 1yr

e.g. what will happen
tomorrow?

w/ daily data; $h=1$

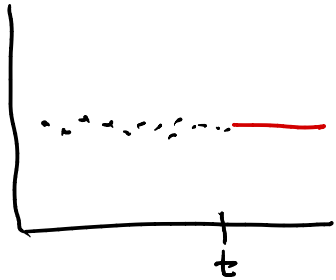
evaluate it w/ cross validation
set $h=1$; step size $=1$

Forecasting techniques:

① Historic Avg

you have data up to time $t \rightarrow y = y_1, y_2, \ldots, y_t$

$$\hat{y}_{t+1|t} = \frac{\sum_{i=1}^{t} y_i}{t}$$



Implications: ① Assumes a stationary time series (it has a fixed mean and variance)

② Leads to a "flat" forecast

↳ your forecast for $\hat{y}_{t+1} = \hat{y}_{t+2} = \ldots$

Note: Practically this approach is hard to defend

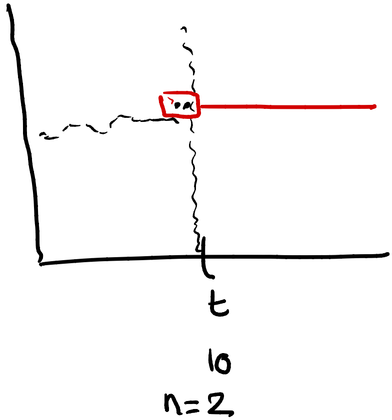
→ if t is Large, then the impact of your Latest obs is minimal

→ that weight is equal for all our data

② Window Average:

$$\hat{y}_{t+1|t} = \frac{\sum_{i=t-n+1}^{t} y_i}{n}$$

assumes stationary data & produces a flat forecast



* Question: Can you evaluate 10 diff versions of these models?

Yes, because I can vary n

* If you are attempting to optimize the choice of "n" $\xrightarrow[\text{that}]{\text{so}}$ you min MAPE or MAE

↳ you will evaluate typically 3 - 10 diff of n

↳ use the value that min your metric

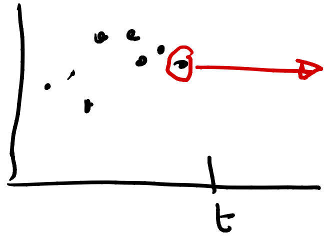
↳ Pick n while cons the freq of your data

* Larger values of n?

↳ will lead to a smoother prediction (i.e. less impact of noise & a smaller Var)

③ Naive forecast

$$\hat{y}_{t+1|t} = y_t$$

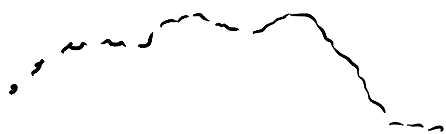


. Produces a flat forecast

• Optimal for a random walk process is mathematically defined using this equation:

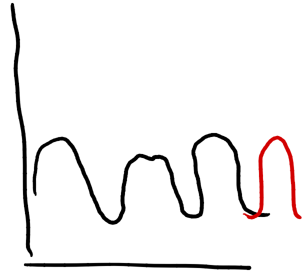
$$\underbrace{y_{t+1}}_{\text{future value}} = \underbrace{y_t}_{\text{current value}} + \underbrace{\varepsilon_t}_{\text{iid error term w 0 mean \& const var}}$$

• random walk model described above is not stationary (mean will be a function of your last obs), but has no trend



④ Seasonal naive model

- no trend
- presence of a strong seasonal component



This does not produce a flat forecast but the pattern will repeat if $h > m$

m → period of data

- annual
  - 4 for quarterly
  - 12 monthly
- daily data
  - 7 for day of week seasonality

5) Seasonal Window Avg

- in the seasonal naive, we use the last Friday to predict next Friday

- we will use the last <u>n</u> Fridays to predict next Friday

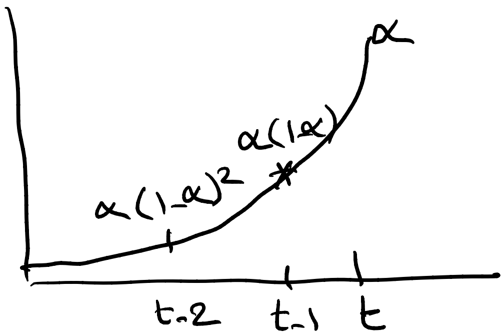
⑥ Simple exponential smoothing method :

Assumes : ① local level can drift slowly over time
(mean written above "level", with an arrow)

② no trend and no seasonality

$$\hat{y}_{t+1|t} = \alpha\, y_t + (1-\alpha)\, \underbrace{\hat{L}_t}_{\hat{y}_{t|t-1}}$$

an equivalent desc

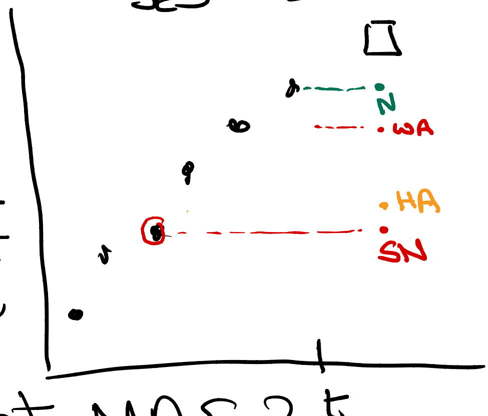


interpretations :

① we are giving more recent data, higher weights in the forecast (weighted avg)

A conceptual question:

None of the prev methods are meant for a ts that exhibits a trend. (HA, $\underset{2}{\text{WA}}$, N, SN, SES) ; $\boxed{n = 4}$



Q1: Assuming you have to pick one of these models to forecast the ts in the plot, which one will have the lowest MAE?

Reminder: error = actual − forecast

$$\varepsilon_t = y_t - \hat{y}_{t|t-1}$$

7 Holts Method

- model that assumes that there is a local trend in the data and the slope can change slightly over time.
- no seasonality
- double exponential smoothing
  - $\alpha \rightarrow$ Level
  - $\beta \rightarrow$ trend

→ This why you are not doing LR (LR assumes a global trend)

8) Holt winters :

↳ as of now, this is the only model that accounts for both a trend & seasonality

Given that this model builds on the idea behind the classical decomp we can have either add or m